

WEEK 4

Title: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

Task 1:

1. Explore Azure Portal

- Create an account on Azure portal

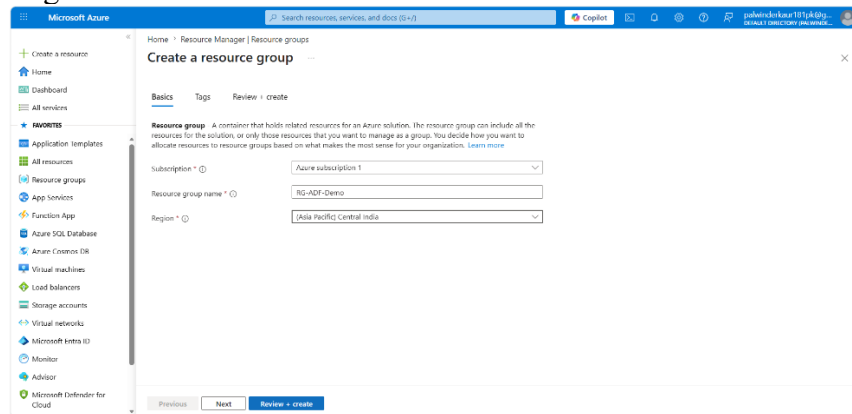
2. Create a Resource Group

Steps :

- a. In the search bar at the top, type: Resource Groups
- b. Click **Resource Groups**.
- c. Click +create and fill the details

Resource Group Name: RG-ADF-Demo

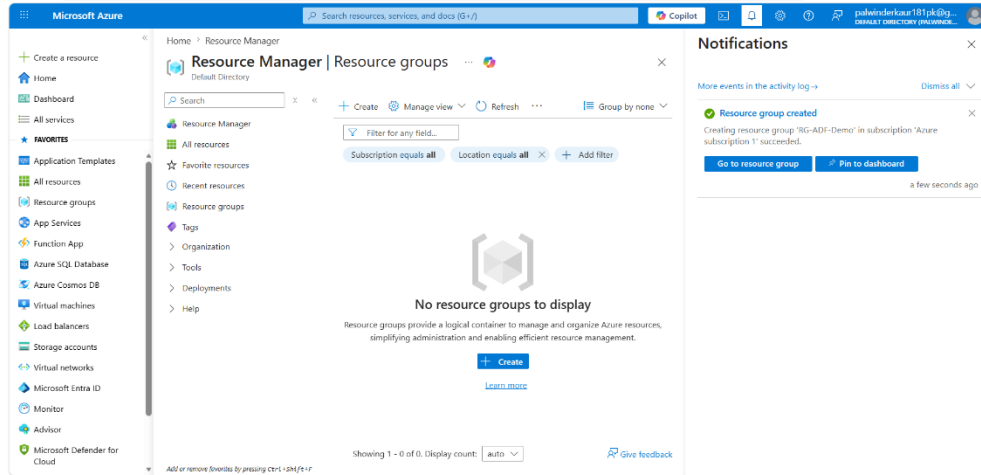
Region: Central India



- d. Click review + create and then click create ,wait for the deployment to complete.

Deliverable:

- Screenshot of Resource Group

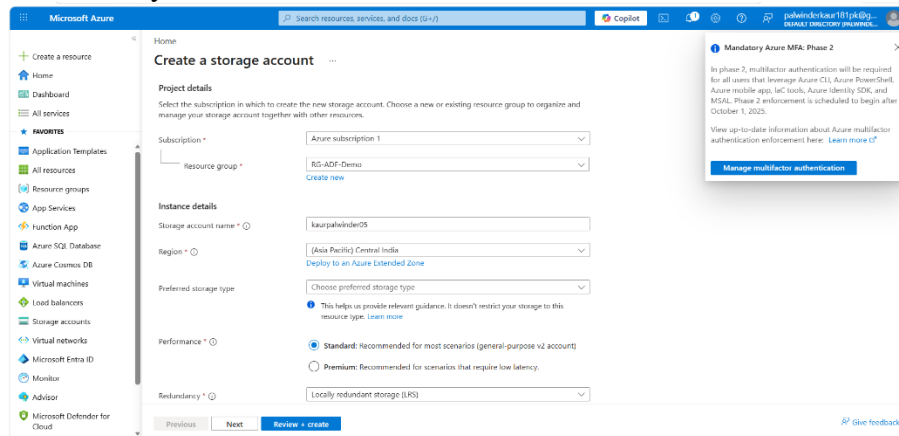


Task 2: Storage Setup

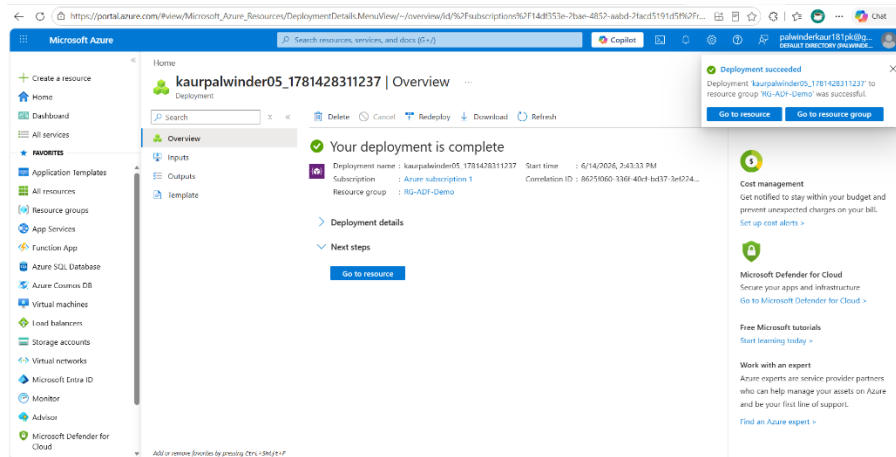
1. Create a Storage Account

Steps to be performed:

- Search Storage Accounts in search bar
- Click + Create and fill the basic information Project Details
Select: Subscription and Resource Group: RG-ADF-Demo
- Fill the Instance Details
Storage Account Name: kaurpalwinder05
- Region: Central India
- Performance: Standard
- Redundancy: Locally Redundant Storage (LRS)

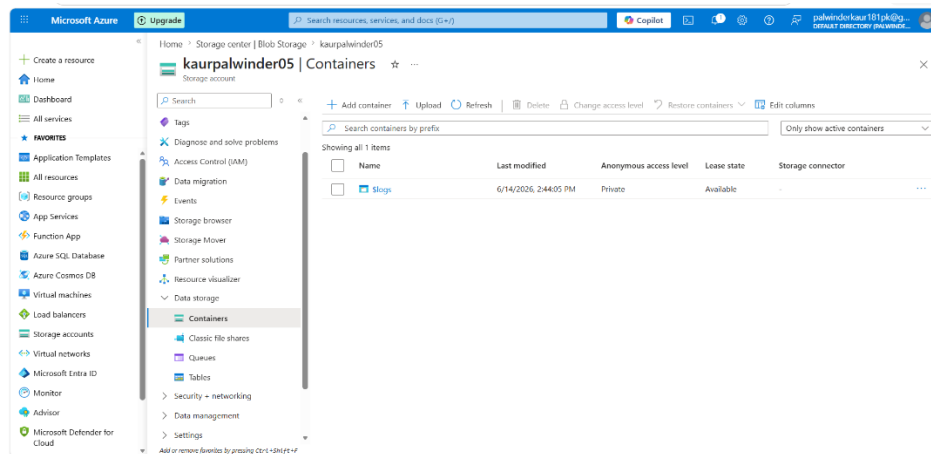


- Click: Review + Create and then click create .
- The deployment takes about 1–2 minutes.



2. Create a Blob Container

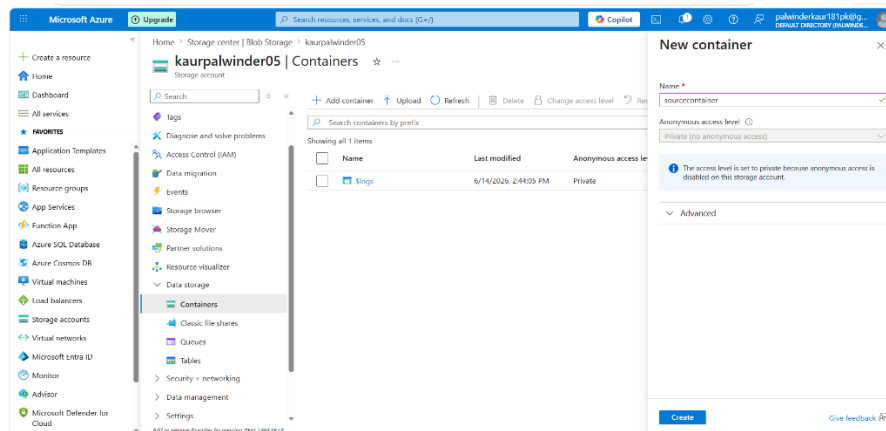
a. Open your Storage Account on its left menu Data Storage → Container.



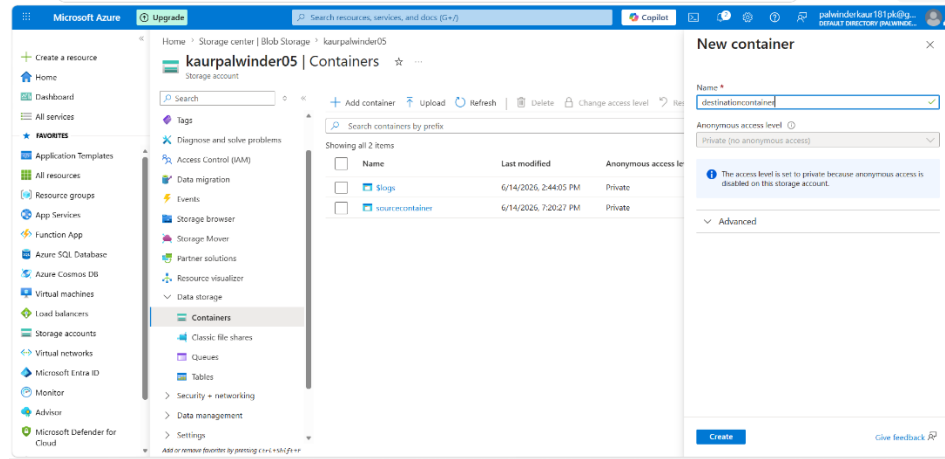
b. Click: + Container

c. Create Source Container

d. Name: sourcecontainer and Click: Create

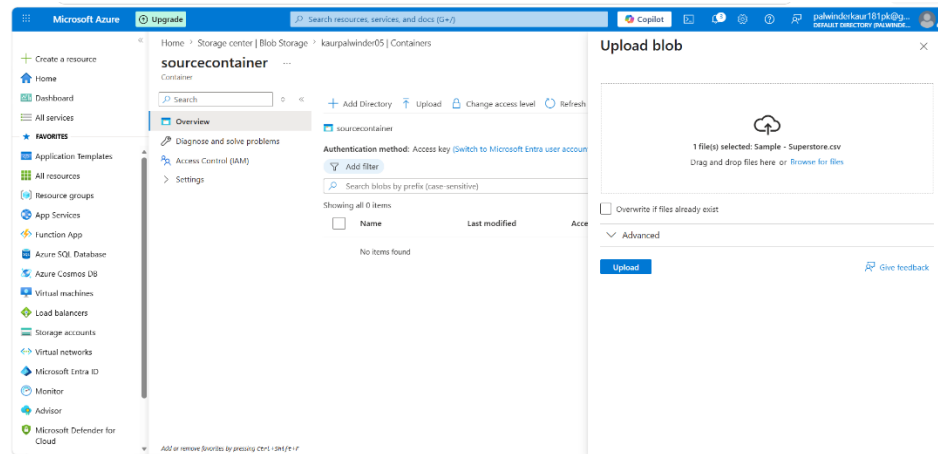


- e. Create Destination Container
- f. Again click: + Container
- g. Name: destinationcontainer and Click: Create



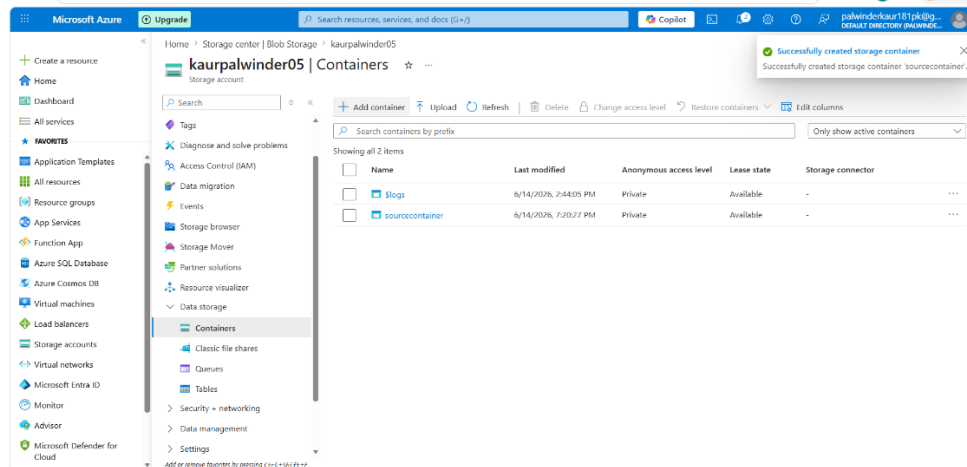
3. Upload a CSV file

Now open sourcecontainer and click: Upload , choose CSV file and click upload.

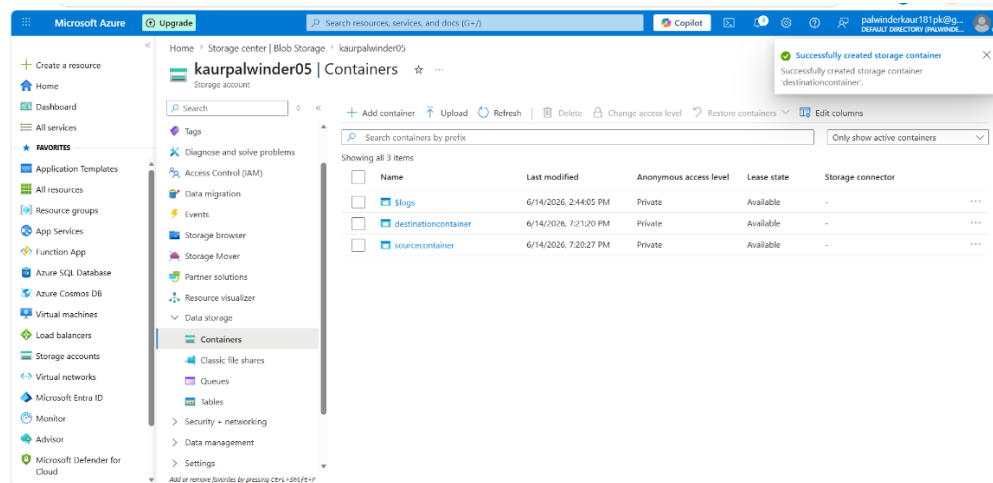


Deliverable:

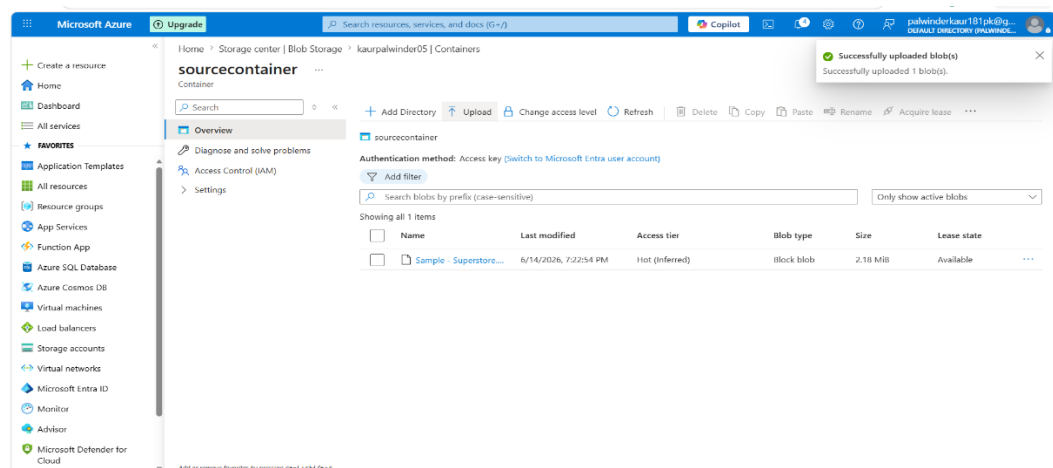
- Screenshot of container with uploaded file
 - a. Screenshot of sourcecontainer successfully created.



b. Screen short of destinationcontainer successfully completed.



c. Screenshot of CSV file successfully uploaded.



Task 3: ADF Basics

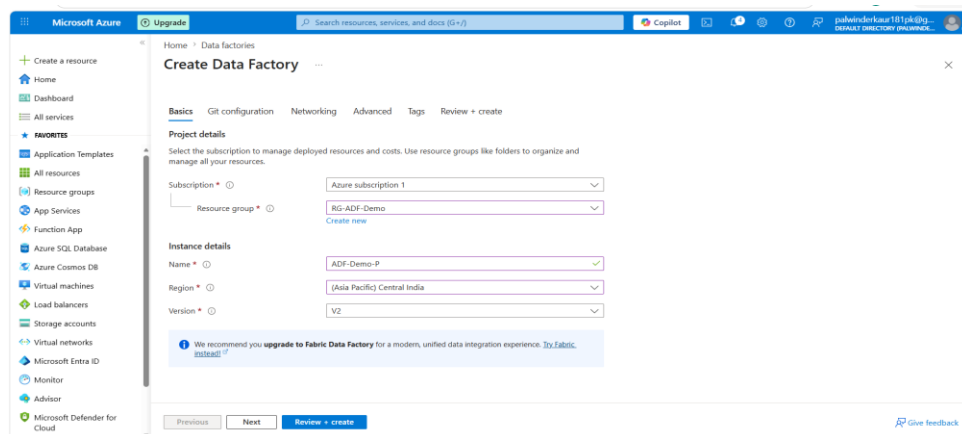
Do:

1. Create Azure Data Factory

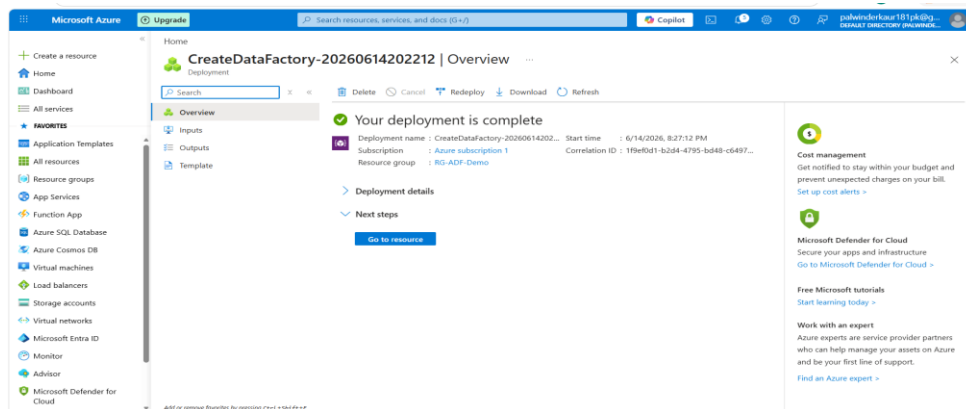
Steps to be performed:

- Search Data Factory in top bar and click + create .
- Fill the details as

Setting	Value
Resource Group	RG-ADF-Demo
Factory Name	ADF-Demo-P
Region	Central India
Version	V2

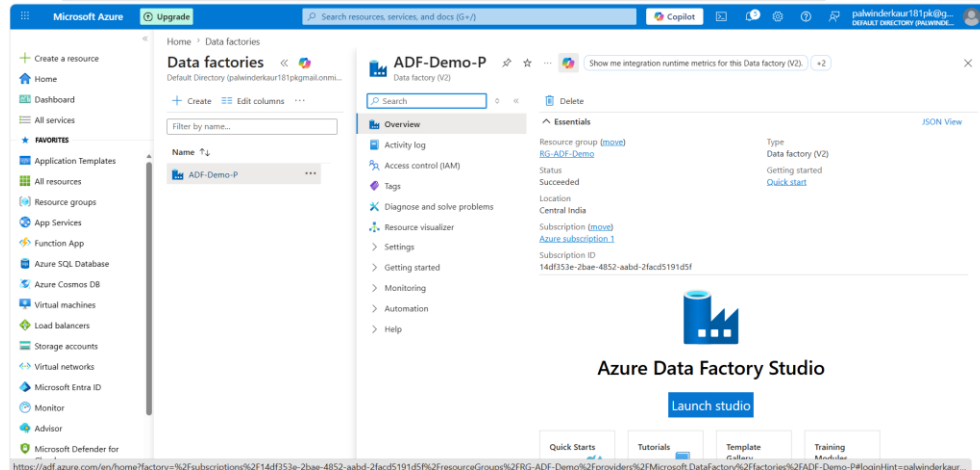


- Click: Review + Create and then click create.
- Deployment takes 2–3 minutes.



2. Explore ADF UI (Author, Monitor, Manage)

a. Open the Data Factory resource click on factory name and click: Launch Studio



We will see three sections:

Author

Used to create:

- Pipelines
- Datasets
- Data flows

Monitor

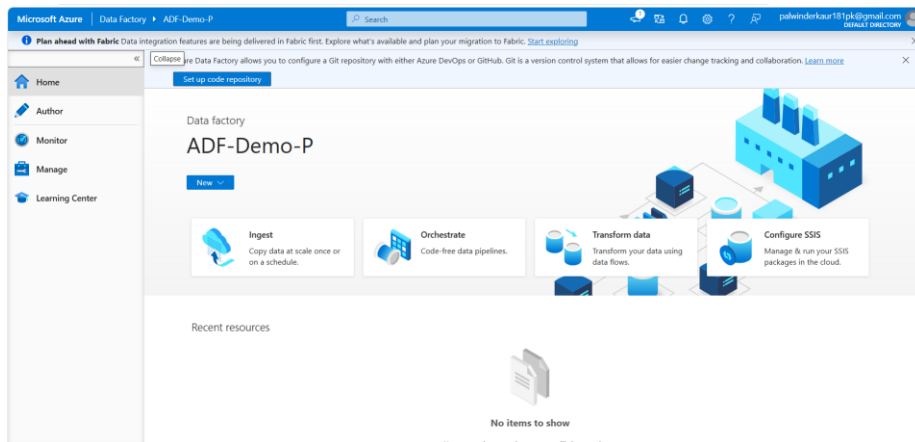
Used to monitor:

- Pipeline runs
- Activity runs
- Success or failure

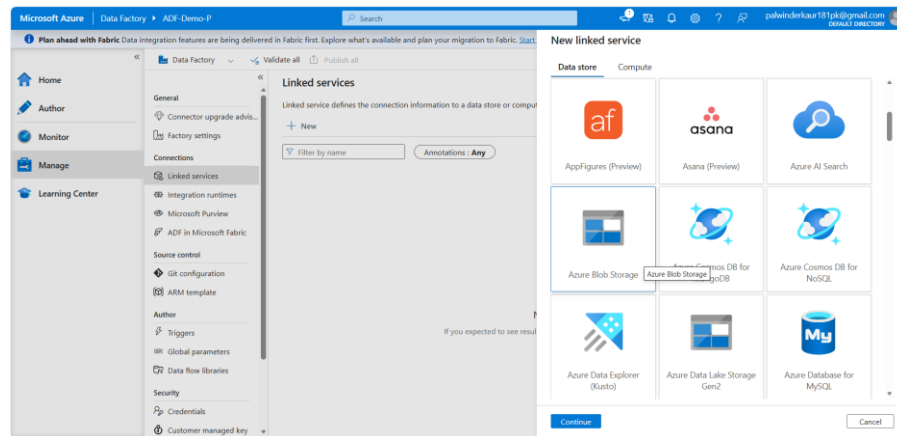
Manage

Used to manage:

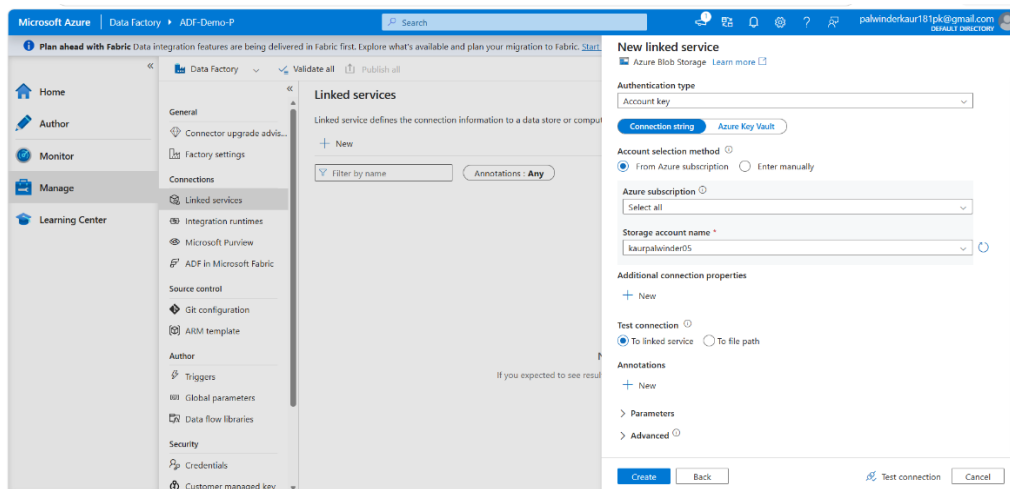
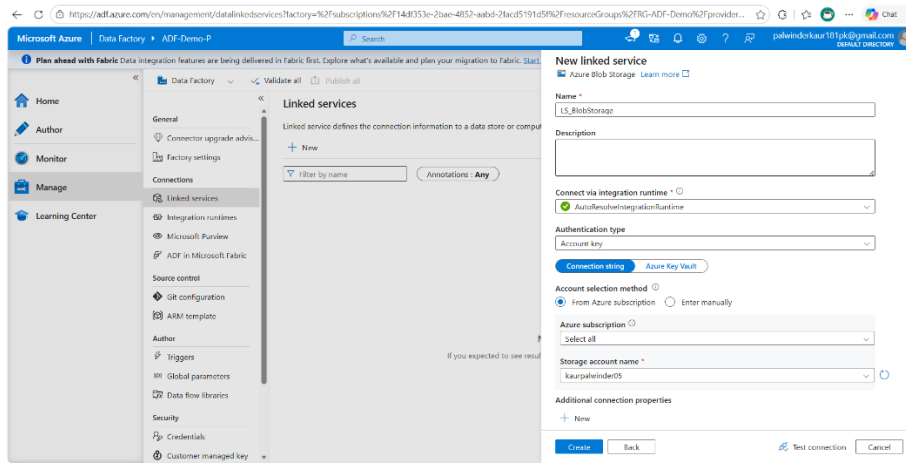
- Linked Services
- Integration Runtime
- Permissions



3. Create Linked Service (Blob Storage)
 - a. Inside the launch studio go to manage→ Linked Services, and click +new.
 - b. Search: Azure Blob Storage, select and click continue.

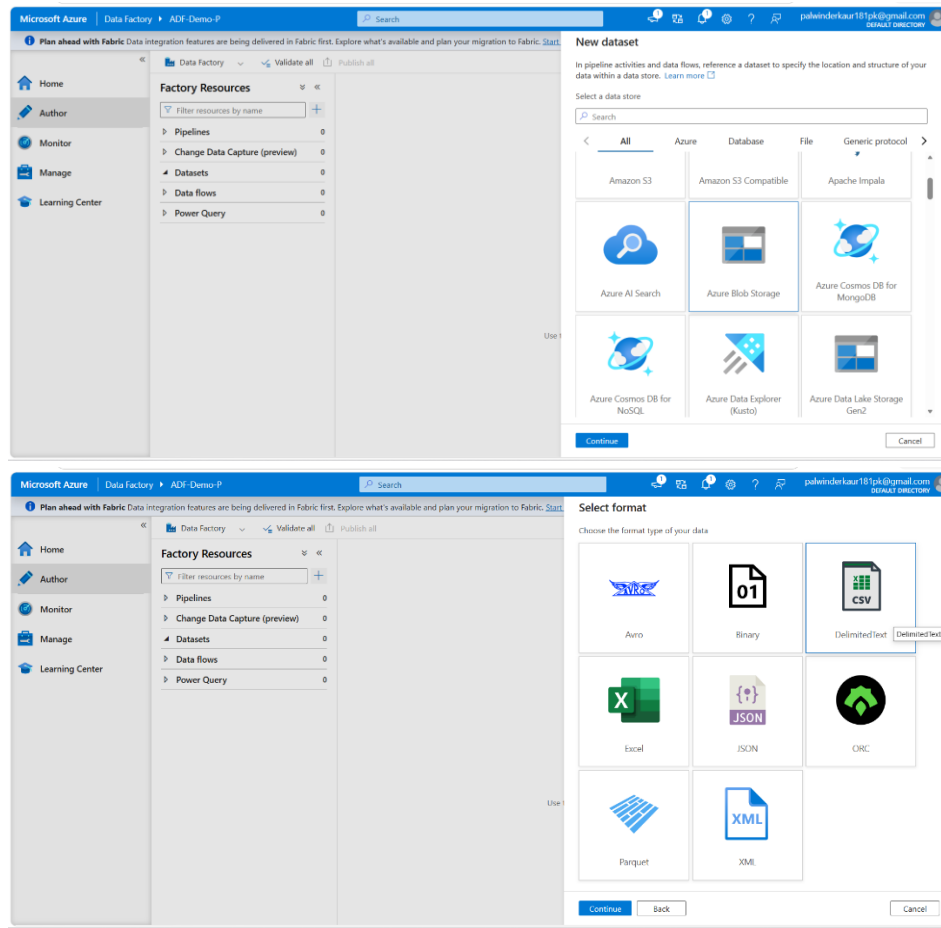


- c. Fill details: Name: LS_BlobStorage , Authentication Method:Account Key, Select: Kaurpalwinder05, Click: Test Connection Connection Successful should appear. Click:Create .

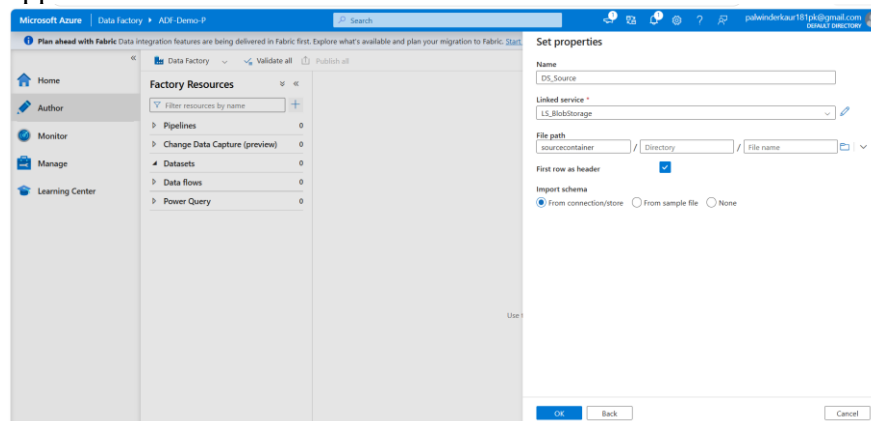


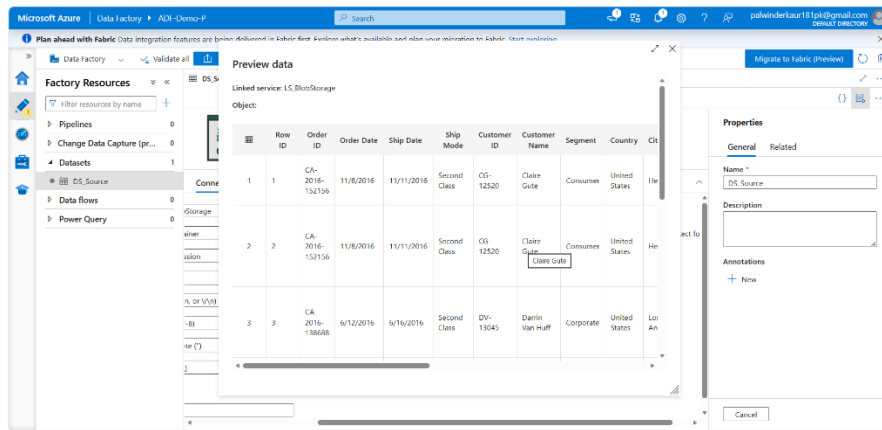
4. Create datasets (source + destination)

- a. Go to Author → +Dataset and choose Azure Blob Storage and select Delimited Text and click continue.

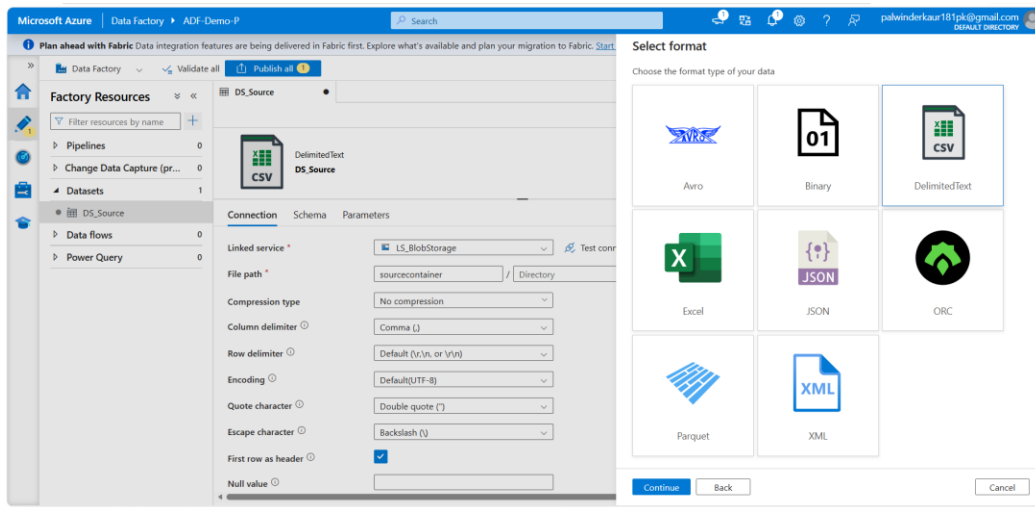


- b. Configure Dataset , Dataset Name: DS_Source, Linked Service: LS_BlobStorage, Container: sourcecontainer, Click ok, Click Preview Data, CSV records should appear.

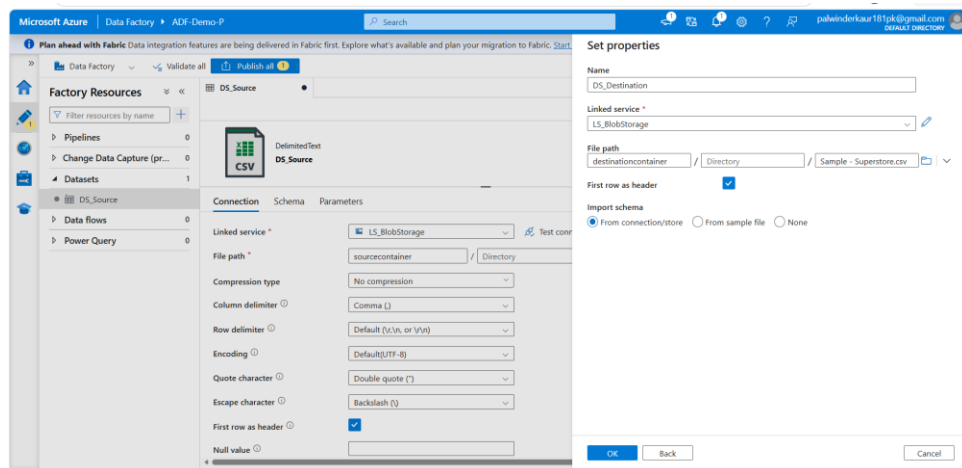




c. Now again: + → Dataset, and choose: Delimited Text.



d. Dataset Name: DS_Destination, Container: destinationcontainer, File Name: filename.csv, Click: OK.



5. Use Get Metadata activity

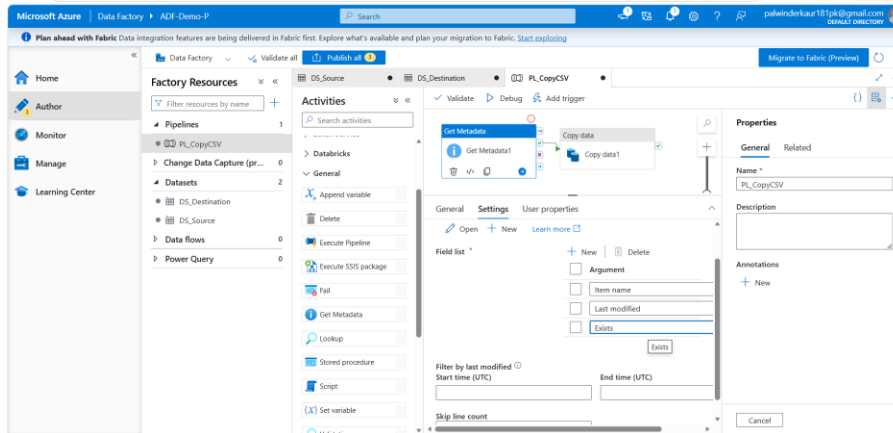
- To create a pipeline go to: Author→+Pipeline.
- Rename: PL_CopyCSV, from Activities panel:drag:Get Metadata to the canvas.
- Open Settings tab select Dataset:DS_Source, Under Field List choose:

✓ File Name

✓ Size

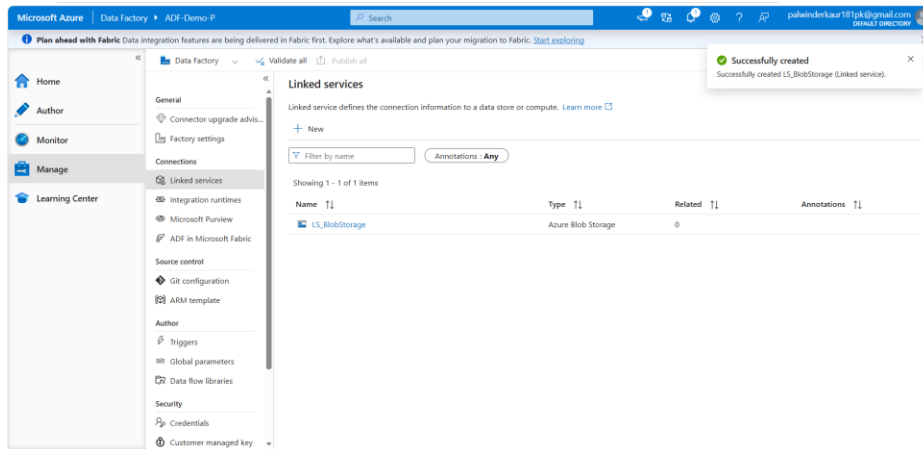
✓ Last Modified

This activity will read information about the file.



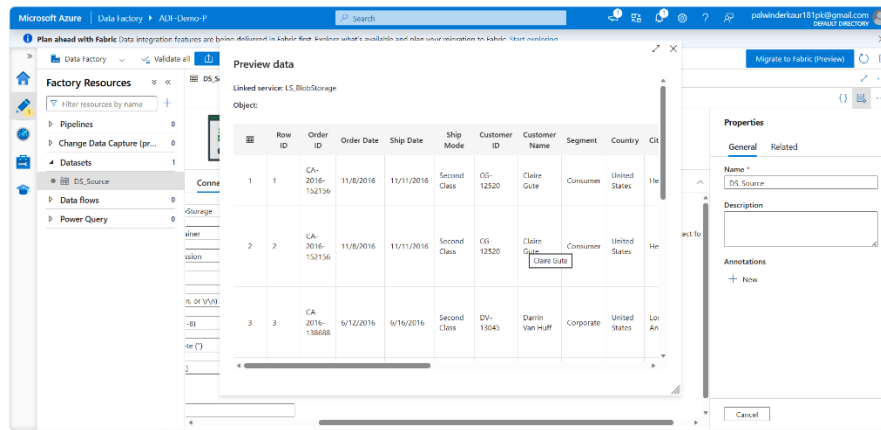
Deliverable:

- Screenshot of Linked Service

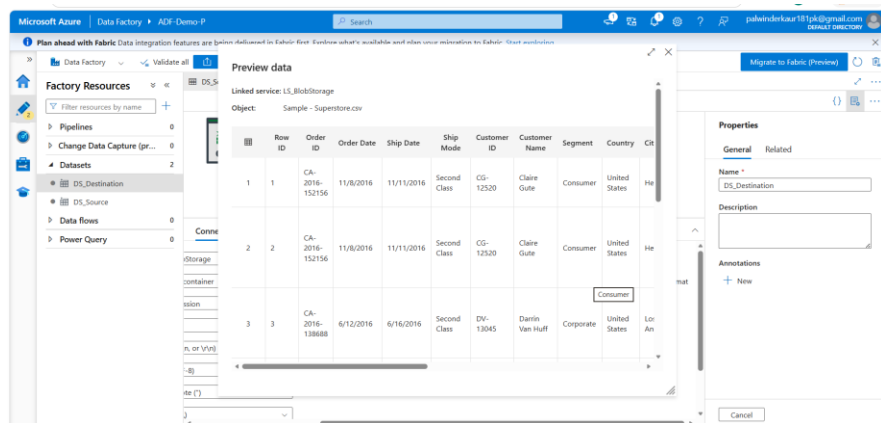


- Screenshot of dataset

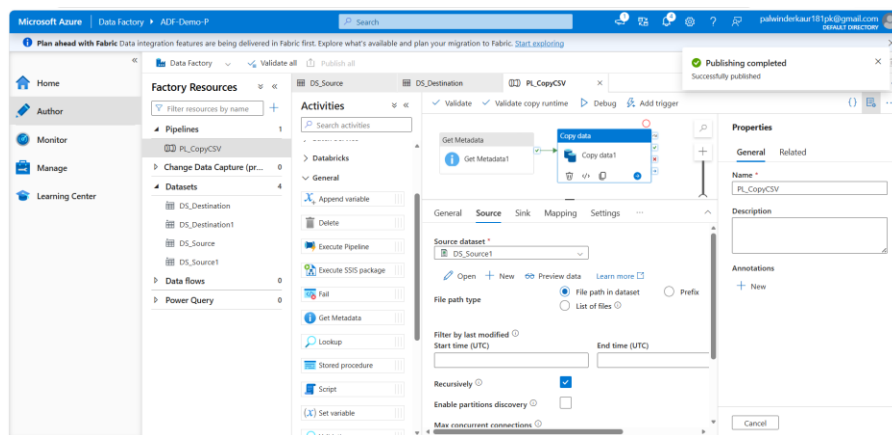
DS_Source Dataset



DS_Destination Dataset:



- Screenshot of Get Metadata activity



Task 4: Pipeline Development

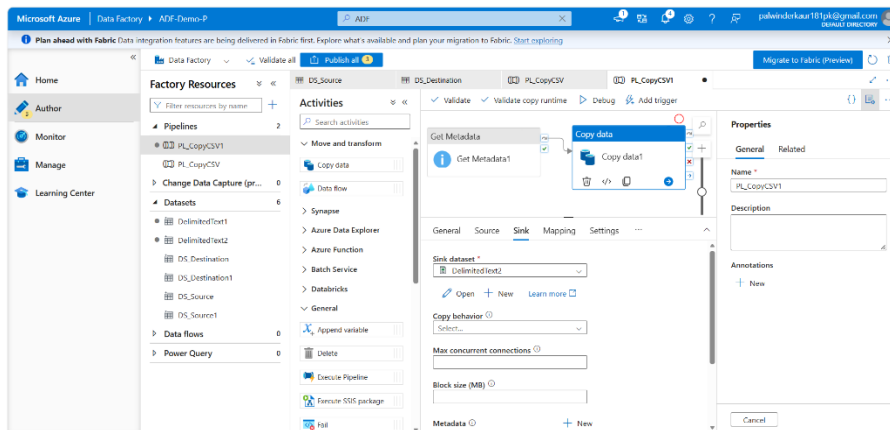
Do:

- Create pipeline using Copy Data activity

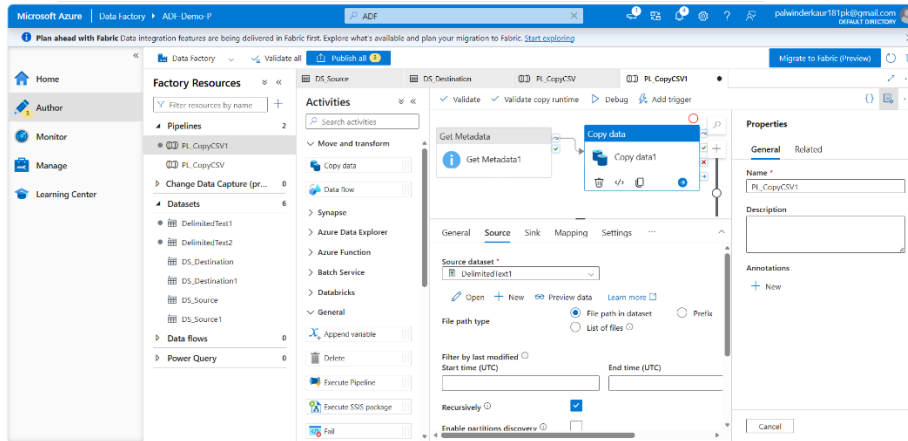
- i. Open **ADF Studio**. Go to **Author** (pencil icon).
 - ii. Click the + button, select: Pipeline, rename the pipeline to: PL_CopyCSV1.
- Configure source and destination
 - a. Click **Copy data1**, Open the **Source** tab, in source dataset, Choose:DS_Source
 - b. In the **Connection** tab:

Property	Value
Linked Service	LS_BlobStorage
Container	sourcecontainer
File	Sample_superstore.csv
 - c. Configure Sink (Destination)
 - d. Click **Copy data1**, Go to the **Sink** tab, Under **Sink dataset**, click:+ New
 - e. Create Destination Dataset, Choose:Azure Blob Storage,DelimitedText→ Continue, General Dataset Name:DS_Destination,Linked Service:LS_BlobStorage,
 - f. Connection Container:destinationcontainer, Click **OK**.
 - g. Now return to **Copy data1** → **Sink tab**, Select:DS_Destination

Sink dataset :

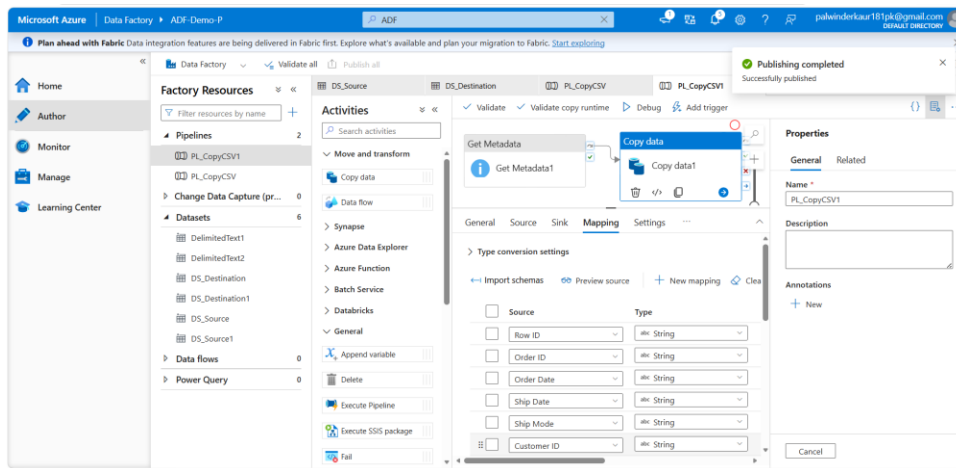


Source Dataset:



Deliverable:

- Screenshot of pipeline design



Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Run pipeline using Debug:-

Step 1: Open ADF Studio,

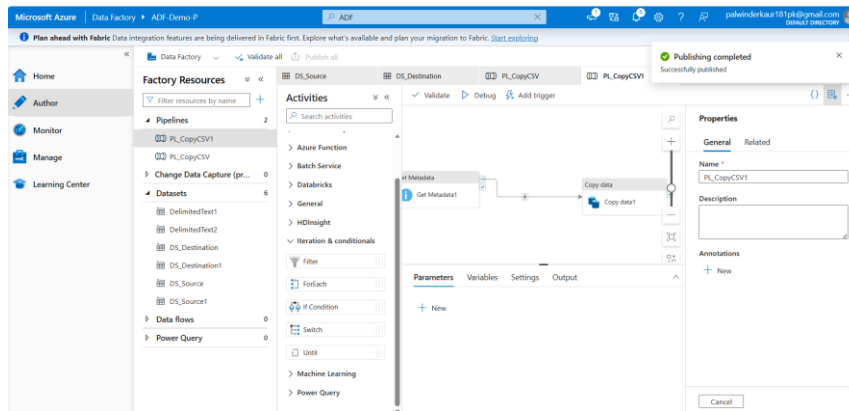
Go to: **Author** (Pencil icon), Open your pipeline: PL_CopyCSV1, we should see: Get Metadata1 → Copy data1.

Step 2: Validate the Pipeline

At the top of the page, click: **Validate**, we should get: No errors found

Step 3: Publish Changes

Click: **Publish All**, wait until: publishing succeeded appears.



Step 4: Click Debug

At the top of the pipeline window, click: Debug, ADF will start executing the pipeline immediately.

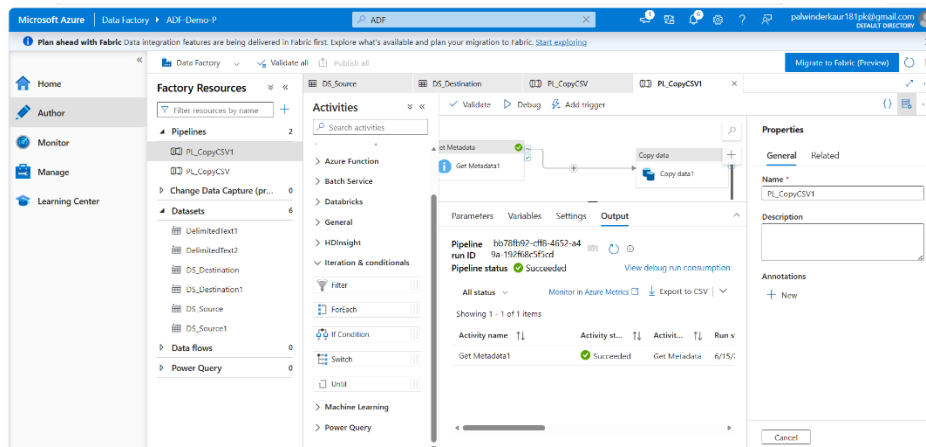
Step 5: Observe Execution Status

At the bottom of the screen, we see: Output- Initially, the status may show:

In Progress, After a few seconds, both activities should turn green.

Deliverable:

- Screenshot showing pipeline execution (Succeeded)



Task 6: IAM Roles

Do:

- Assign roles: Reader
 - Open Your Storage Account, select Storage Accounts:kaurpalwinder05.
 - From the left menu,Access Control (IAM), Click it.
 - Add Reader Role, at the top, click: + Add,from the dropdown menu, select: add role assignment.Choose Reader Role.

- iv. Under the Job function roles tab:search:Reader, select:Reader, Click: Next.
- v. Select Members, we will reach the Members page.Under: Assign access to Choose: Managed identity, Then click:+ Select members.
- vi. Select Your Data Factory,a panel will open on the right, subscription choose your Azure subscription, Managed Identity from the dropdown select:Data Factory,Select Choose your ADF.
- vii. Click: select,Assign Reader Role, Click: Review + assign, Again click: Review + assign, Reader role is now assigned.

Assign Contributor Role

- i. Access Control (IAM) →+ Add →Add role assignment
- ii. Search:Contributor, Select:Contributor,Click:Next.
- iii. Select Members- Choose:Managed identity, Click:+ Select members.
- iv. Subscription - Select your subscription, Managed Identity, Choose:Data Factory, Select Choose:ADF-Demo,Click:Select.
- v. Assign Contributor Role- Click:Review + assign,Again click:Review + assign, Contributor role has now been assigned.

• Provide access to ADF for Storage

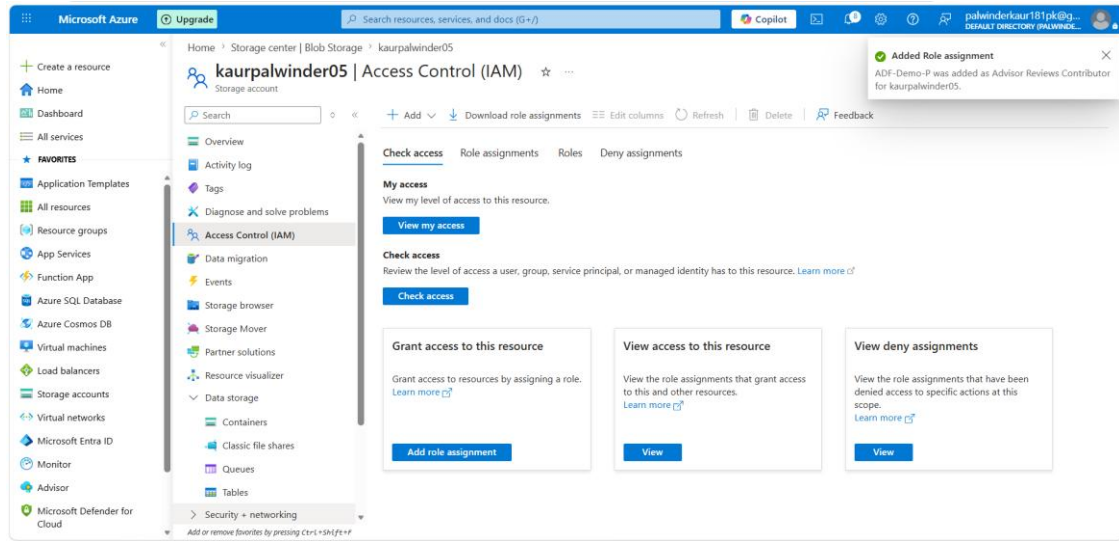
- vi. Verify Role Assignments- Inside:Storage Account→Access Control (IAM)→Role Assignments, Search for:ADF-Demo.

Role	Assigned To
Reader	ADF-Demo
Contributor	ADF-Demo

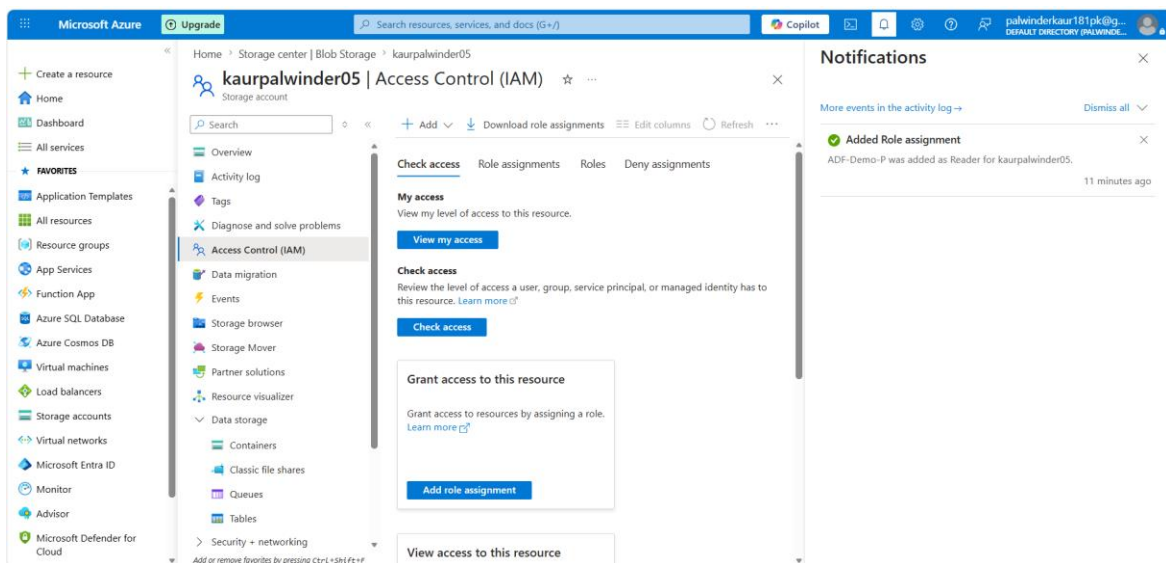
Deliverable:

- Screenshot of role assignment

Contributor role assigned:



Reader role assigned:



Mini Project

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

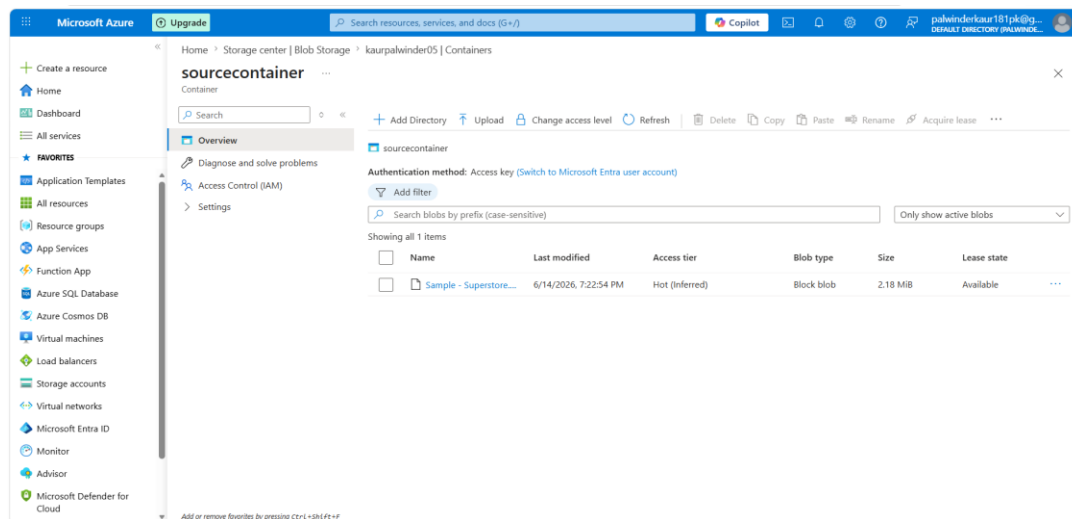
Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Steps to perform:

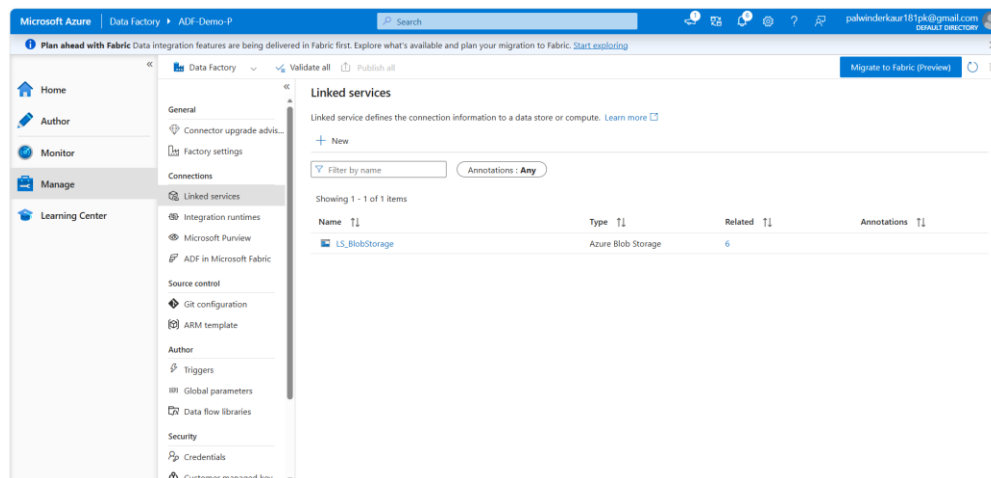
Step 1: Verify Source File

Go to: Storage Account → Containers → sourcecontainer, make sure your CSV file exists.



Step 2: Verify Linked Service

Open: ADF Studio → Manage → Linked Services, ensure: LS_BlobStorage exists and the connection test is successful.



Step 3: Verify Source Dataset

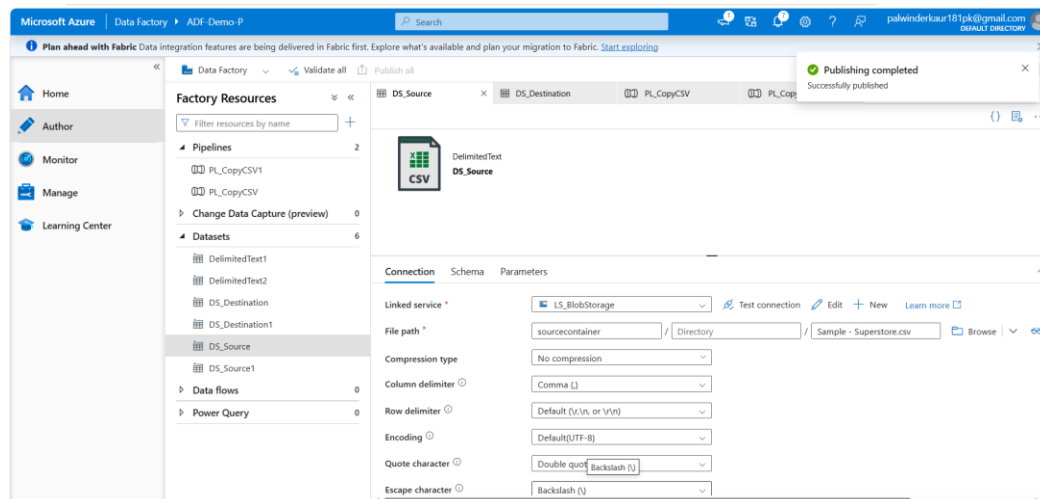
Go to: Author → Datasets → DS_Source, Under Connection, verify:

Property Value

Container sourcecontainer

File Sample-Superstore.csv

Click: Preview Data, to confirm the file contents.



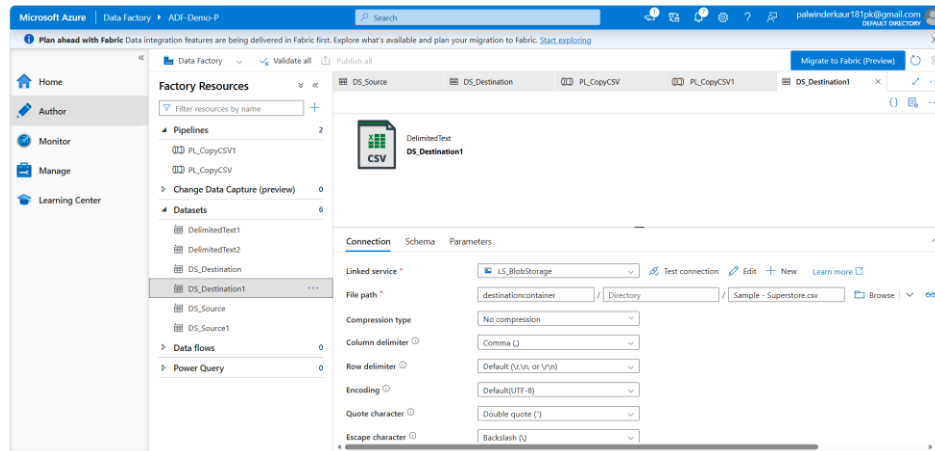
Step 4: Verify Destination Dataset

Go to: Author → Datasets → DS_Destination, Verify:

Property Value

Container destinationcontainer

File Sample-Superstore.csv



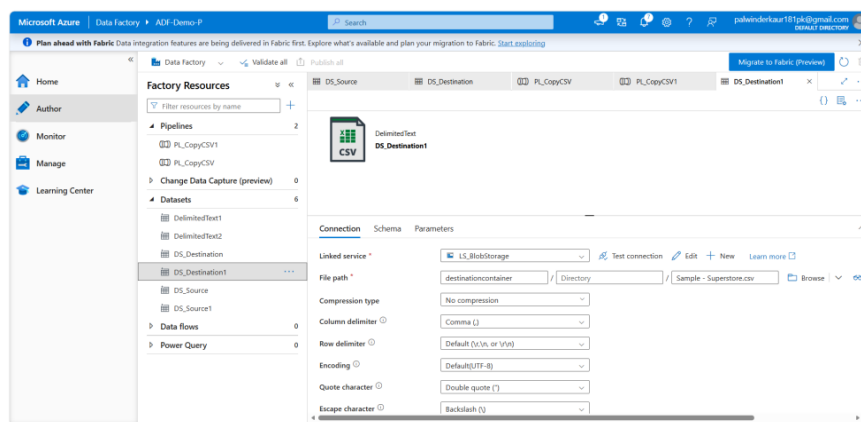
Step 5: Create Pipeline

Go to: Author → + New Pipeline, Rename it: pipeline_Superstore.

Step 6: Add Get Metadata Activity

From Activities: General → Get Metadata, drag it to the canvas.

Configure Get Metadata- **Settings** → Dataset: DS_Source, Field List: Add: Item name, Last modified, Exists.



Step 7: Add Copy Data Activity

From Activities: Move & Transform → Copy Data, drag it onto the canvas.

Step 8: Connect Activities

Connect: Get Metadata1 → Copy data1, using the green arrow.

Step 9: Configure Source and Sink

Click **Copy data1**, Open the **Source** tab, Dataset: DS_Source.

Open the **Sink** tab, Dataset:DS_Destination, Destination file: sample-Superstore.csv

Step 10: Configure Mapping

Open:Mapping, Click: Import Schemas, ADF automatically maps columns.

Step 11: Validate Pipeline

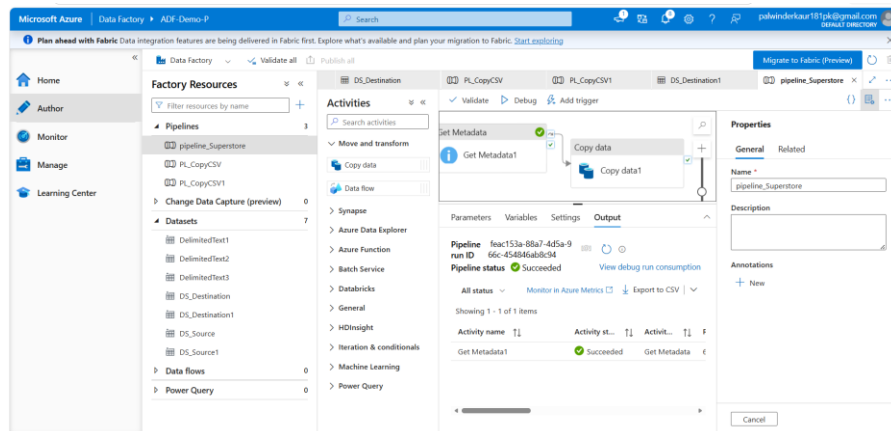
Click:Validate, Expected message: No errors found.

Step 12: Publish Pipeline

Click: Publish All, Wait until: Publishing succeeded appears.

Step 13: Execute Pipeline

Click:Debug.



Step 15: Monitor Execution

Open:Monitor, You should see:

Pipeline Name	Status
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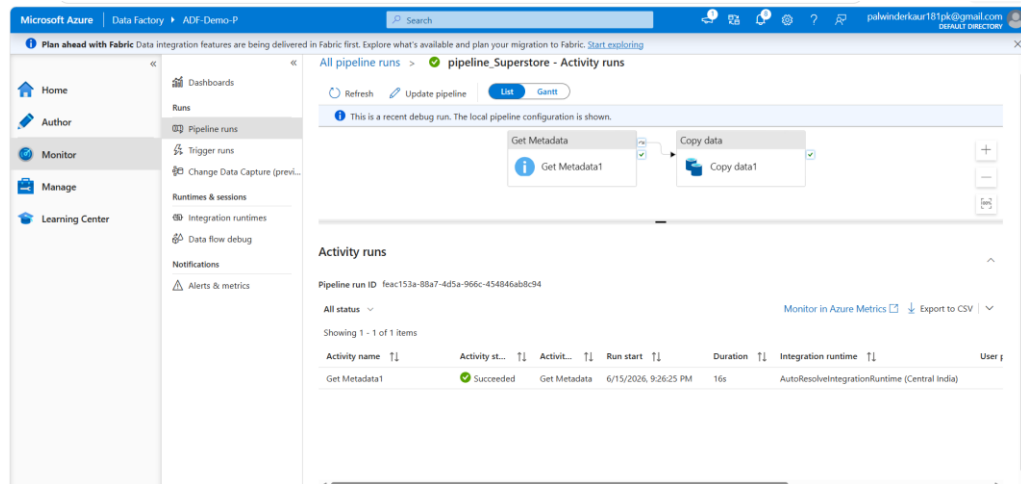
Pipeline_Superstore	Succeeded
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Click the pipeline run, Activity status should show:

Activity Status

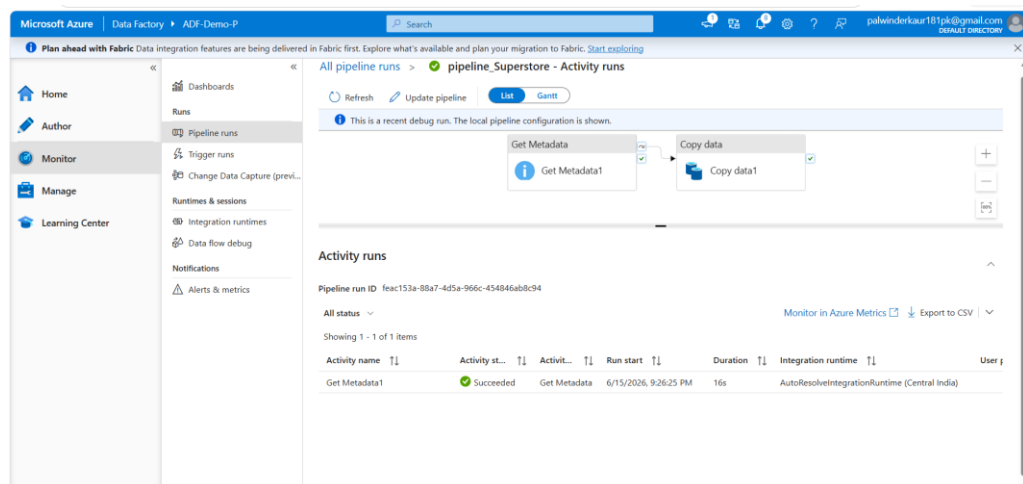
Get Metadata1 Succeeded

Copy data1 Succeeded

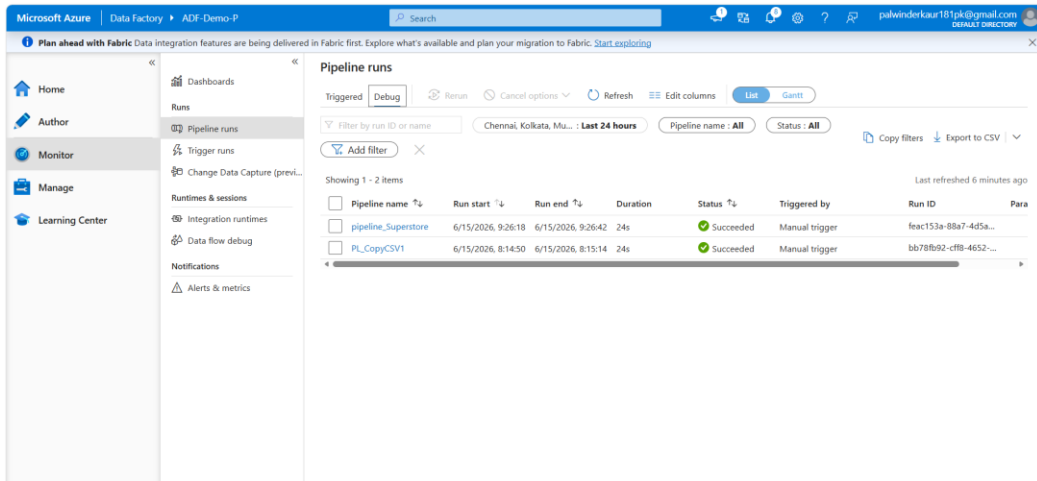


Expected Output:

- Pipeline executed successfully



- Data copied to destination



- Metadata validated

Open: Monitor
 ↓
 Pipeline Run
 ↓
 Get Metadata
 ↓
 Output

